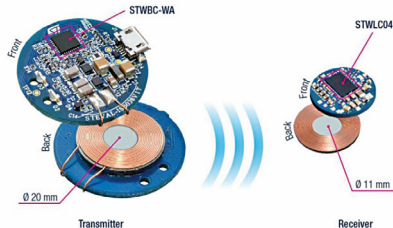


WIRELESS CHARGERS

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Wireless Battery Charger



This reference design is for a wireless battery charger optimised for efficient 1-watt wireless power transfer. The design features Qi 1.0 standard and is specifically designed for wearable devices and IoT applications.

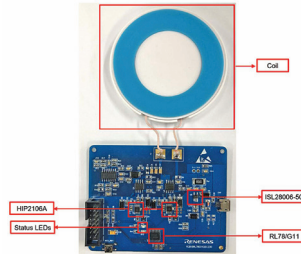
The STEVAL-ISB038V1 is a reference design by STMicroelectronics, offering an optimised wireless battery charger for efficient 1-watt wireless power transfer. The design utilises a half-bridge topology on the transmitter side and incorporates compact 11mm and 20mm coils on the receiver and transmitter sides, respectively. The transmitter features a powerful software application programming interface (API) for customisation. The design is specifically designed to facilitate a 1-watt power transfer utilising the Qi protocol. The design incorporates a combination of digital and precise analogue control loops to ensure stable and reliable operation. The design features an inter-integrated circuit (I2C) interface, offering extensive customisation options for users. The reference design follows the Qi 1.0 standard and operates with disabled foreign object detection (FOD) as its low output power does not necessitate it. The system maintains stable performance even when there are foreign objects present or when transmitters and receivers are in motion. Incorporating wireless power transfer technologies, the design offers extensive integration capabilities, resulting in compact and cost-effective designs with reduced time-to-market.

The reference design is suitable for wearable gear, smartwatches, Internet of Things (IoT) sensors, and healthcare devices.

STMicroelectronics

<https://www.electronicsforu.com/electronics-projects/electronics-design-guides/reference-design-of-a-wireless-battery-charger>

Wireless Charging Transmitter



This reference design for a wireless charging transmitter features Qi standard. The design operates at 5V and within a temperature range of -10°C to 45°C. The maximum charging current is 800mA, and the maximum output power is 4W.

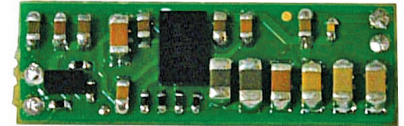
The RL78/G11 from Renesas is a reference design for a wireless charging transmitter. The reference design features Qi wireless charging standard. The design functions on a 5V DC power supply, with an operating temperature range of -10°C to 45°C and operating humidity range of 5% to 99% RH. The wireless charger design provides a maximum charging current of 800mA and a maximum output power of 4 watts. The wireless charging transmitter consists of several components such as a DC 5V power adaptor, a current sense amplifier, a voltage detection circuit, a temperature monitor circuit, a main MCU, an LED indication circuit, a PWM driver circuit, and a coil. The micropower current sense amplifier enables efficient operation with its low power consumption and wide input voltage range. The power transmission process will only initiate if the power supply voltage falls within the acceptable range of 4.5V to 5.5V, which is monitored by the voltage detection circuit. Additionally, the design incorporates a temperature monitoring circuit to continuously measure the temperature of the transmitter coil. If the temperature exceeds the established safety threshold of 60°C, the power transmission will automatically be halted.

The reference design is suitable for creating smartphone chargers, automotive applications, etc.

Renesas

<https://www.electronicsforu.com/electronics-projects/electronics-design-guides/reference-design-for-wireless-charging-transmitter>

Wireless Charger for Wearable Applications



This reference design for a wireless charger follows Wireless Power Consortium (WPC) standards and is Qi certified. The design can handle charging current in the range of 10mA to 250mA. The design boasts a maximum leakage current of 75nA from 0°C to 125°C.

The TIDA-00318 is a wireless charger from Texas Instruments (TI). The design follows Wireless Power Consortium (WPC) standards for establishing wireless communication and is Qi certified. The reference design effectively handles charging currents ranging from 10mA to 250mA, supporting termination currents as low as 1mA. Specifically tailored for applications that require a charge current of 135mA, the design of the wireless charger is compact and optimised. The wireless receiver incorporates rectification, voltage conditioning, and controller blocks to enable efficient wireless charging. Combining the wireless receiver design with a wireless transmitter IC creates a seamless and complete charging solution. The design showcases an impressive maximum battery leakage current specification of 75nA, which remains consistent across a wide temperature range from 0°C to 125°C. To provide some context, at this current, it would take around 152.2 years for a 100mAh battery to fully discharge. In reality, this current is negligible since the self-discharge rate of the battery cell surpasses it by a significant margin.

The reference design is suitable for low-power wearable devices. The design is optimised for applications requiring a charge current of 135mA. It can be used in various industrial applications like in a position sensor, etc.

Texas Instruments (TI)

<https://www.electronicsforu.com/electronics-projects/industry-grade-projects/reference-design-for-a-wireless-charger-for-wearable-applications>